Mutual of Omaha

DQC, R&S

Senior Design Showcase

April 24,2026

Company

- **Mutual of Omaha**
  - Insurance and financial services company
  - Headquartered in Omaha, Nebraska since 1909
  - One of Nebraska's few Fortune 500 companies
  - Known for their strong local community involvement
- **What They Do**
  - Provide financial protection products and services
  - Focus on meeting the unique needs of their customers
- **The Challenge**
  - Manage large, complex collections of business data
  - Existing data processes lacked automation and reliability
  - A software solution was needed to ensure data quality and integrity

Rendering of the new Mutual of Omaha headquarters being built in Omaha Nebraska

Stakeholders

Sponsors

Jerrid Derr - I/S Director

Chris Poore - I/S Manager

Sai Chillara - Architect II

Krishna Ravipati - Engineer

Pete Sheffield - Architect IV

Adam McQuistan - Engineer IV

Scott Zenor - I/S Director

Stakeholders

Faculty, Staff and Coach

Project Manager
Bill Browning

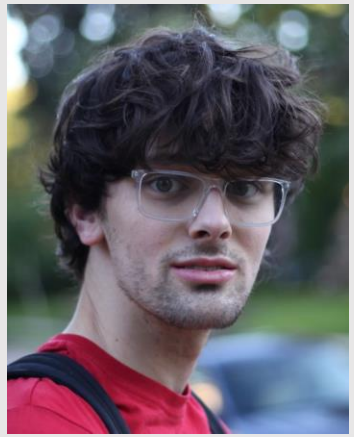
Tribe Lead
Gurvinder Singh

Coach
Steve Arnold

Stakeholders

Student Team

Darius Banks – Product Manager



Matthew Rokusek - Developer

Aditya Tadepalli - Developer

Ceferino Patino – Development Manager

Levi Loseke - Developer

Jaden Davis - Developer

Problem and Opportunity

- Data storage in JSON documents
  - Defined by Schemas
  - Data has become inconsistent and contains errors
  - Hinders business applications
  - Slows down development
- Complex Relationships
  - Relational vs Non-relational
  - Complexity has increased overtime
- Demands a software solution to address these concerns
  - Data quality and referential integrity validation
  - Reporting

```
[
  {
    "Id": 0,
    "FirstName": "string",
    "LastName": "string",
    "Name": "string",
    "EmailAddress": "string",
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  }
]
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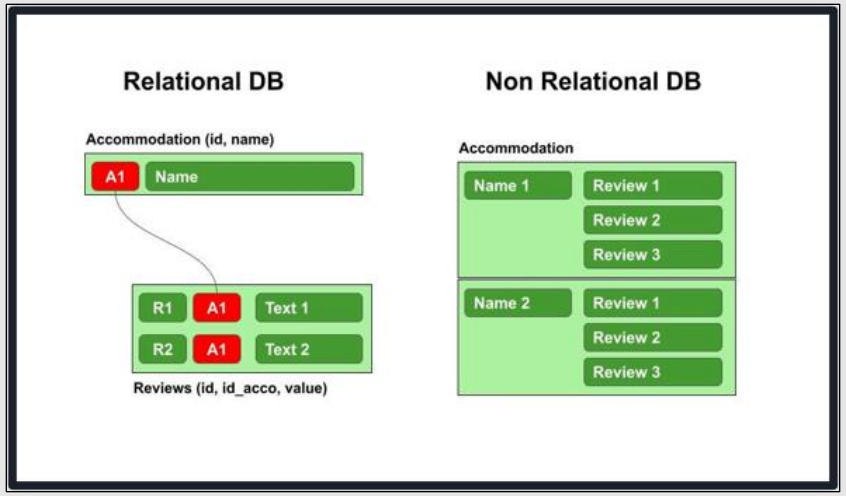
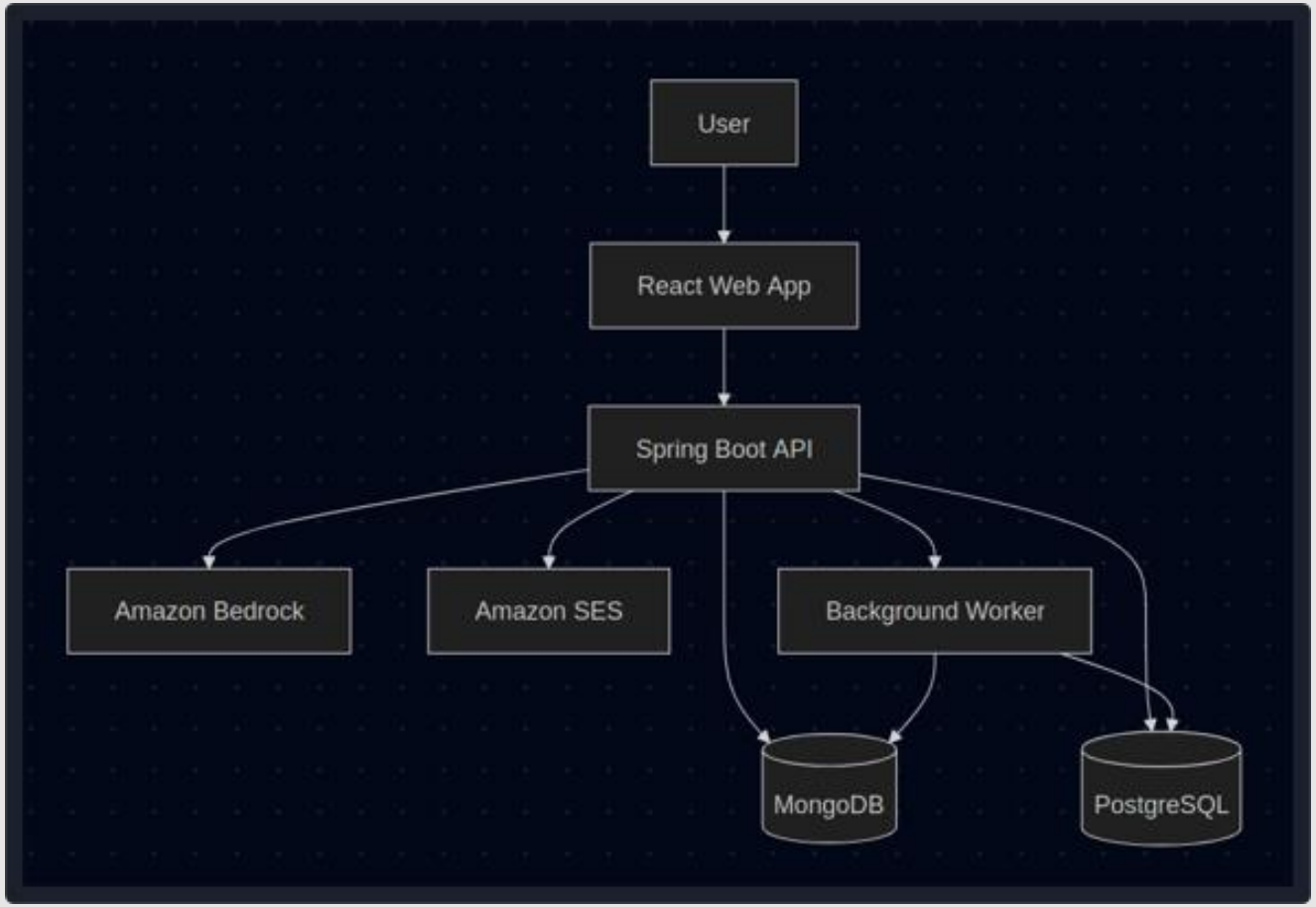


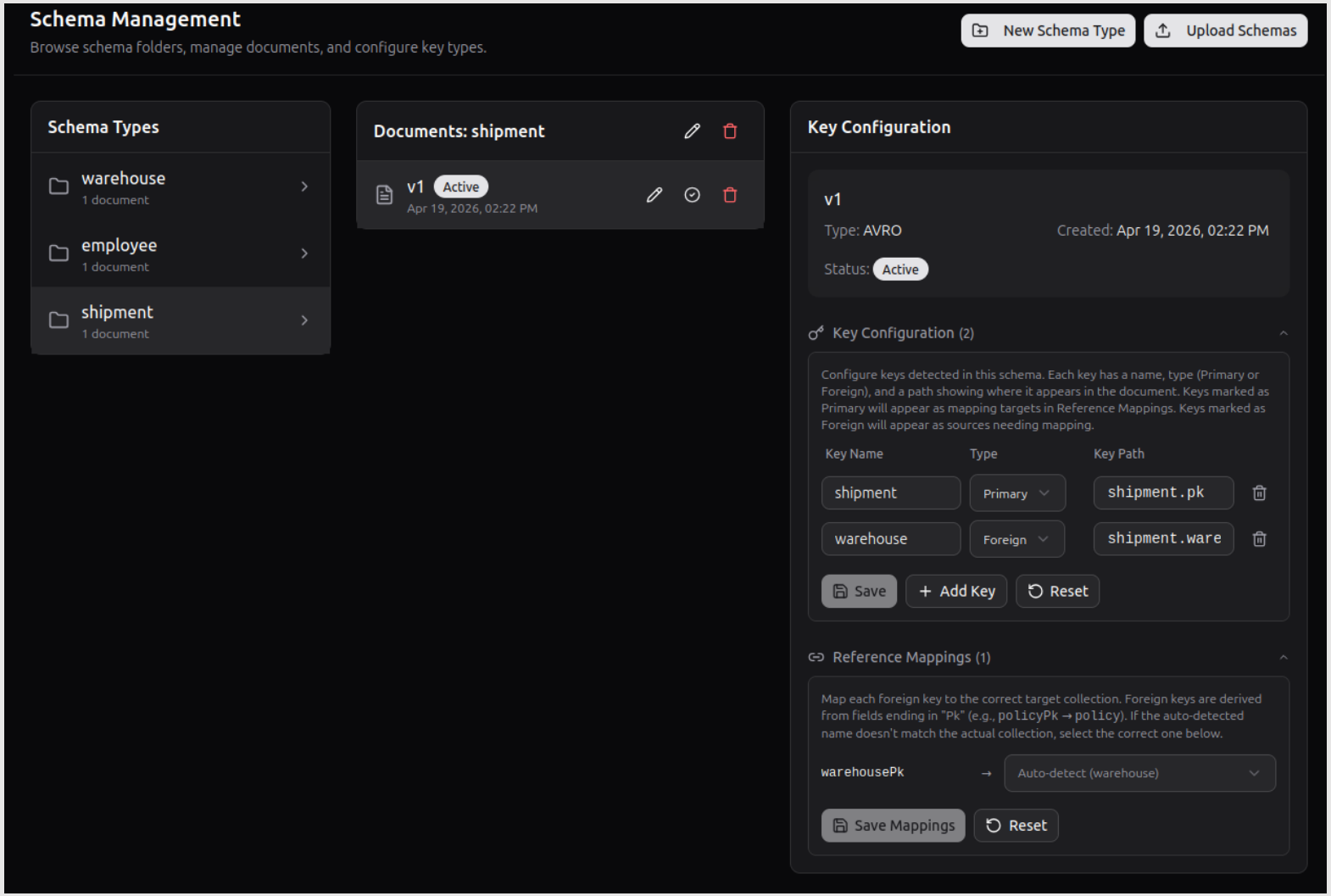
Product Solution

- Our application includes:
  - JSON document validation based on defined schemas
  - Referential integrity checks among related documents
  - Historical validation report viewing with document exporting
  - Monitoring service with email notifications
  - AI report summaries



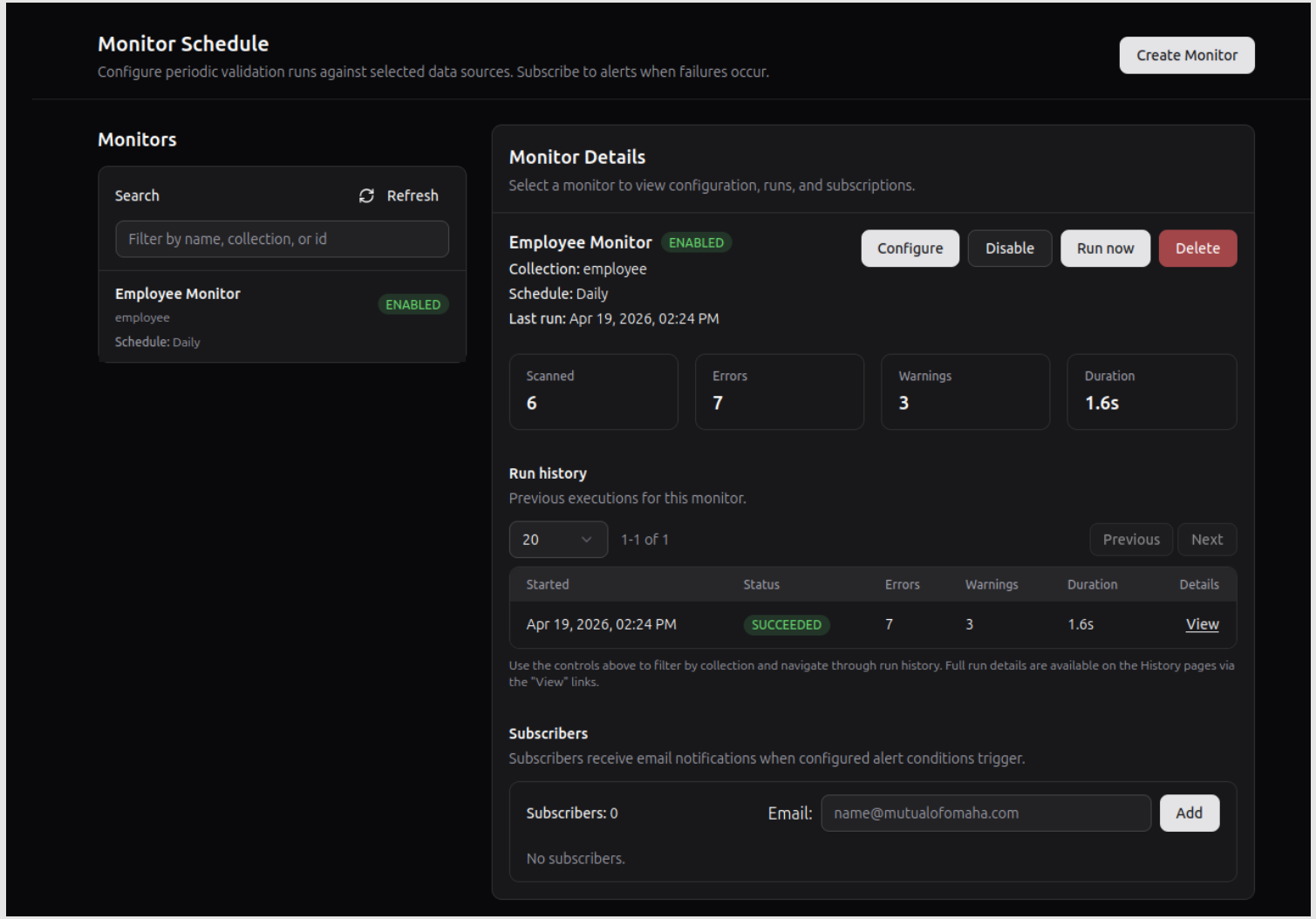
Product Solution

- Schema Management
  - Schema versions can be uploaded/managed
  - Keys can be configured
  - Foreign key relationships can be customized



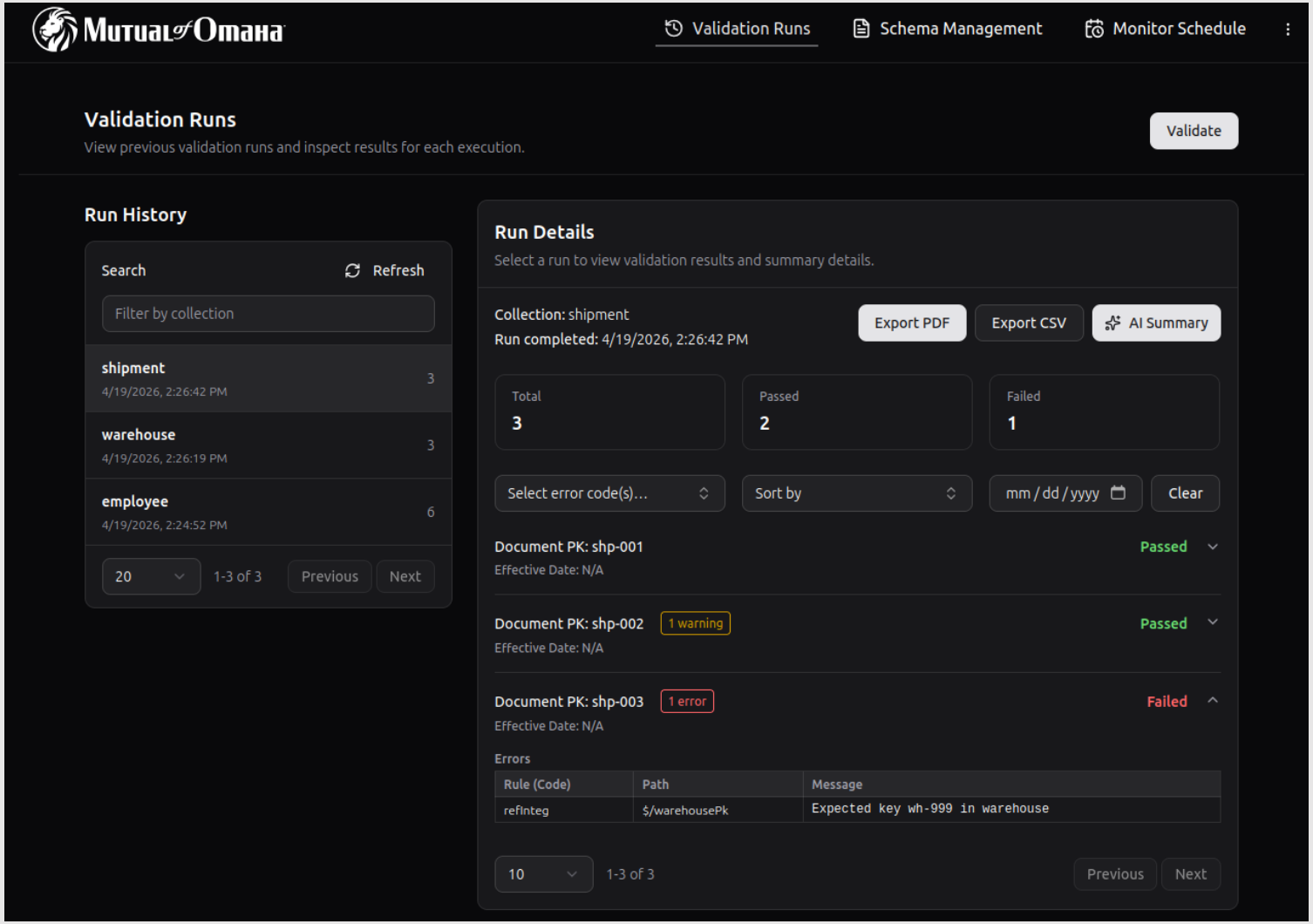
Product Solution

- Monitor Creation
  - Monitors can be created/configured
  - Email notifications can be set
  - Runs can be manually run



Product Solution

- Validation Runs
  - Previous runs can be viewed and filtered/sorted
  - Reports can be exported
  - AI summaries can be generated



Value and Highlights

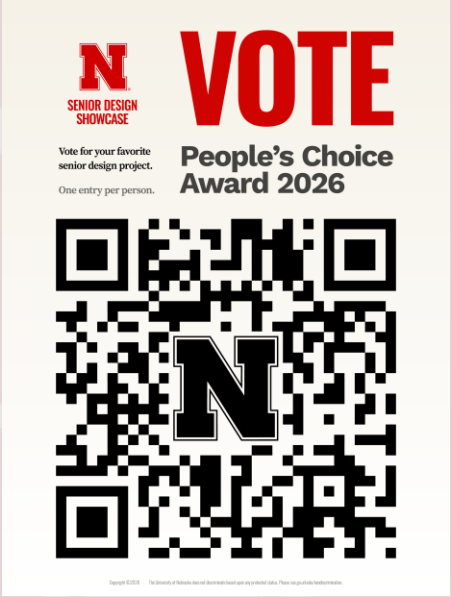
Value Provided:

- Could reduce the time teams spend manually tracking down data inconsistencies.
- The monitoring and alerting system could allow issues to be caught earlier, preventing costly downstream errors.
- As their data grows, the platform could scale to support additional schemas and collections at scale.

What We Learned:

- Enterprise problems rarely have clean solutions, so we had to develop a plan for a problem where it wasn't known what the solution would look like.
- Communication is critical in a large organization, and decisions require more coordination and patience than in smaller teams.
- Planning ahead is essential when working through admin and IT departments. Waiting on approvals or access can stall a project, so getting ahead of those processes early kept us on track

Thank You!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